T03 (A2): selection, screening, PIPs and the global HPM

SMC-BVS v3 target, regime R1

Proof by the wave-3 lead (Claude Opus 5.5); not adopted

22 October 2026 (version 2: exposition revisions after review round 1; no mathematical step changed)

Status: proof submitted for review; *not adopted*. It proves the statement in `STATEMENT.md`, with T02(i) as the explicit hypothesis H-T02. All notation, hypotheses and conclusions are those of `STATEMENT.md`; they are recalled where used. Nothing else is assumed beyond standard linear algebra and probability; the two chi-square facts used are proved in Lemma 7.

Contents

1 Setting and plan **1**

2 Reduction to contrast coordinates **2**

3 Deterministic preliminaries **3**

4 The good event **5**

5 Supersets of size at most m_n **6**

6 Underfitting models of size at most m_n **7**

7 Proof of the theorem **8**

8 Remarks **10**

1 Setting and plan

Recall: $n \geq 3$, $N := n-1$; standardised data $X = CX^rD_x^{-1}$, $y = Cy^r/s_y$; $K = \min(p, \lfloor n/2 \rfloor)$, $\Omega = \{S : |S| \leq K\}$; $a_0 = b_0 = 0.01$, $\nu = a_0 + n/2$; $w(k) = \frac{1}{2}/\binom{2p-1}{k}$; $\pi(S) \propto w(|S|)m(S)$ on Ω; $r_n = \sqrt{\log n/n}$; $m_n = \lceil (s_0 \log(2p) + (s_0+2)(\log n)/2)/\log 2 \rceil$ (this is $m_n(\frac{1}{2}\log n)$). Hypotheses G, R, SE, BM-C, BM-∞, H-T02 and the constants j_*, C_*, L_*, M_* are as in `STATEMENT.md`. We also use

$$C_A := 8 + 3A, \qquad \tau_j := C_A j \log n \ (j \geq 1).$$

For a model S (any $S \subseteq [p]$, inside Ω or not) the formula for $m(S)$ makes sense, and we write $\ell(S) := w(|S|)\, m(S)$. If $S_0 \in \Omega$ then $Z \geq \varpi(S_0)m(S_0)$ and, for every $S \in \Omega$,

$$\pi(S) \leq \frac{\pi(S)}{\pi(S_0)} = \frac{\ell(S)}{\ell(S_0)}. \tag{1}$$

For a model S put $J(S) = S_0 \setminus S$, $J'(S) = S \setminus S_0$, $S' = S \cup S_0$. S is a *superset* if $J(S) = \emptyset \neq J'(S)$, and *underfitting* if $J(S) \neq \emptyset$.

Asymptotic conventions. n_0 denotes a threshold depending only on $(s_0, A, \varphi_0, \sigma_\star^2, M)$, enlarged finitely many times; "for $n \geq n_0$" means for all n beyond it. $X_n = O_{P_0}(a_n)$ means that X_n/a_n is tight under P_0. Since $p \leq n^A$ and $\log 2 \leq \log n$, $1 \leq \log n$ for $n \geq 3$,

$$m_n \leq C_m \log n, \qquad C_m := \frac{s_0(1+A) + (s_0+2)/2}{\log 2} + 1. \tag{2}$$

By R, $s_0 \leq K$ for $n \geq n_0$; in Sections 4–7 we take $n \geq n_0$ unless said otherwise, so $S_0 \in \Omega$ and (1) holds. The deterministic sequences $\delta_n, \epsilon_n, q_n$ defined below depend on the design only through m_n and are nondecreasing in m_n; replacing m_n by $C_m \log n$ from (2) gives dominating sequences that depend only on $(s_0, A, \varphi_0, \sigma_\star^2, M)$ and n. This is why every threshold of the form "$\delta_n \leq \frac{1}{2}$", "$q_n \leq \frac{1}{2}$", "$q_n \leq \sqrt{eC_A}\, r_n$" can be placed in n_0. (Under BM-∞, one extra threshold depending on the sequence (M_n) enters in Step 1 of Section 7; all conclusions concern one sequence of problems, so this is harmless.) Notation: $w(\cdot)$ is always the prior weight function; the contrast response vector is y_c.

Plan. Section 2 rewrites the target in contrast coordinates, where the data are exactly Gaussian with N coordinates and the response scale enters only through the inverse-gamma rate $b = b_0 s_y^2$. Section 3 collects deterministic facts. Section 4 defines a good event of probability $1 - O(1/n)$. On it, Section 5 bounds the supersets of size at most m_n, one size at a time, and Section 6 bounds the underfitting models of size at most m_n by comparing each S with $S' = S \cup S_0$. Models larger than m_n are handled by H-T02. Section 7 assembles (i)–(v).

2 Reduction to contrast coordinates

Let $U \in \mathbb{R}^{n \times N}$ with $U^\top U = I_N$ and $UU^\top = C$. Put

$$X_c := U^\top X \in \mathbb{R}^{N \times p}, \qquad y_c := U^\top y^r \in \mathbb{R}^N, \qquad e := U^\top \varepsilon, \qquad \mu_c := X_c \theta^\star.$$

For $S \subseteq [p]$ let $X_{c,S}$ be the columns of X_c in S, P_S the orthogonal projection of $\mathbb{R}^N$ onto $\mathrm{col}(X_{c,S})$ ($P_\emptyset = 0$), and

$$M_S := n\, X_{c,S} H_S^{-1} X_{c,S}^\top \ (M_\emptyset = 0), \qquad t_S := y_c^\top M_S y_c, \qquad Q_S^c := \|y_c\|^2 - t_S, \qquad b := b_0 s_y^2.$$

Lemma 1 (reduction)**.** *For every valid input:*

(a) $G_S = X_{c,S}^\top X_{c,S}$, $s_y z_S = X_{c,S}^\top y_c$, $\|y_c\|^2 = N s_y^2$ *and* $s_y^2 Q_S = Q_S^c \geq 0$. *Consequently, for all* $S, T \subseteq [p]$,

$$\log \frac{\ell(S)}{\ell(T)} = \log \frac{w(|S|)}{w(|T|)} - \frac{1}{2} \log \frac{D_S}{D_T} - \nu \log \frac{2b + Q_S^c}{2b + Q_T^c}. \tag{3}$$

(b) $y_c = \mu_c + e$ *and, under* G, $e \sim N_N(0, \sigma_\star^2 I_N)$. *Moreover* $\|\mu_c\| = \|\mu\|$, *and* $\mu_c \in \mathrm{col}(X_{c,S_0})$, *so* $(I - P_S)\mu_c = 0$ *whenever* $S \supseteq S_0$.

Proof. (a) C is a symmetric idempotent and $X = CX^r D_x^{-1}$, so $CX = X$. Hence $X_c^\top X_c = X^\top UU^\top X = X^\top CX = X^\top X$, which gives the first identity. Next $s_y z_S = X_S^\top C y^r = X_S^\top UU^\top y^r = X_{c,S}^\top y_c$, and $\|y_c\|^2 = (y^r)^\top UU^\top y^r = (y^r)^\top C y^r = N s_y^2$, which also equals $s_y^2\, y^\top y$. Therefore

$s_y^2 Q_S = \|y_c\|^2 - n(s_y z_S)^\top H_S^{-1}(s_y z_S) = Q_S^c$. Nonnegativity: by Lemma 2(b) below, $M_S \preceq P_S \preceq I$. Since $2b_0 + Q_S = s_y^{-2}(2b + Q_S^c)$,

$$\log m(S) = \log \frac{(2b_0)^{a_0}\Gamma(\nu)}{\pi^{n/2}\Gamma(a_0)} + \nu \log s_y^2 - \frac{1}{2}\log D_S - \nu \log(2b + Q_S^c),$$

and the first two terms do not depend on S. This gives (3).

(b) $\|U^\top \mathbf{1}\|^2 = \mathbf{1}^\top C \mathbf{1} = 0$, so $U^\top \mathbf{1} = 0$. Also $U^\top C = U^\top U U^\top = U^\top$ and $CX^r = XD_x$, so $U^\top X^r \beta^\star = U^\top C X^r \beta^\star = U^\top X D_x \beta^\star = X_c \theta^\star$. Hence $y_c = U^\top(\alpha^\star \mathbf{1} + X^r\beta^\star + \varepsilon) = \mu_c + e$, and $e = U^\top \varepsilon \sim N_N(0, \sigma_\star^2 U^\top U) = N_N(0, \sigma_\star^2 I_N)$ under G. Further $\|\mu_c\|^2 = \theta^{\star\top} X^\top C X \theta^\star = \|X\theta^\star\|^2 = \|\mu\|^2$, and $\mu_c = X_{c,S_0}\theta^\star_{S_0}$ since $\operatorname{supp}\theta^\star = \operatorname{supp}\beta^\star = S_0$ (D_x is diagonal with positive entries). □

Under G the set of ε for which y^r is constant has probability zero ($n \geq 2$, $\sigma_\star^2 > 0$), so π is P_0-almost surely defined; it is a measurable function of ε. From now on everything is expressed through (X_c, y_c, e, μ_c), (3) and (1). The design quantities G_S, H_S, D_S, P_S, M_S are deterministic; b, t_S, Q_S^c are random. Note that b is the same for every model, so it never has to be integrated against; it only needs to be bounded (Lemma 10(a)).

3 Deterministic preliminaries

Write $h(\lambda) = n\lambda/((n+1)\lambda + 1)$ for $\lambda \geq 0$, and

$$\eta_n := \frac{1}{(n+1)N\varphi_0^2}.$$

Lemma 2 (ridge smoother)**.** *Let $S \neq \emptyset$.*

(a) $M_S = h(X_{c,S}X_{c,S}^\top)$ (functional calculus). Hence M_S vanishes on $\operatorname{col}(X_{c,S})^\perp$, and on $\operatorname{col}(X_{c,S})$ its eigenvalues are $h(\lambda)$ for the nonzero eigenvalues λ of G_S.

(b) $0 \preceq M_S \preceq \frac{n}{n+1}P_S \preceq P_S$.

(c) If $\lambda_{\min}(G_S) \geq N\varphi_0^2$, then $M_S \succeq (1-\eta_n)\frac{n}{n+1}P_S$.

(d) If $S \subseteq T$ then $M_S \preceq M_T$; hence $t_S \leq t_T$. (Also true for $S = \emptyset$.)

Proof. (a) With $a = n+1$ and $Y = X_{c,S}$, $Y(aY^\top Y + I) = (aYY^\top + I)Y$, so $Y(aY^\top Y + I)^{-1} = (aYY^\top + I)^{-1}Y$ and $M_S = n(aYY^\top + I)^{-1}YY^\top = h(YY^\top)$, the two factors commuting. The nonzero eigenvalues of $YY^\top$ are those of $Y^\top Y = G_S$, and their eigenvectors span $\operatorname{col}(Y)$; $h(0) = 0$. (b) h is increasing with $0 \leq h < \frac{n}{n+1}$; apply (a). (c) On $\operatorname{col}(Y)$ every eigenvalue of $YY^\top$ is $\geq N\varphi_0^2$, so by (a) and monotonicity $M_S \succeq h(N\varphi_0^2)P_S$, and $h(g) = \frac{n}{n+1}(1 - \frac{1}{(n+1)g+1}) \geq \frac{n}{n+1}(1 - \frac{1}{(n+1)g})$; take $g = N\varphi_0^2$. (d) $h(\lambda) = \frac{n}{n+1}(1 - ((n+1)\lambda+1)^{-1})$. If $0 \preceq B_1 \preceq B_2$, then $((n+1)B_2 + I)^{-1} \preceq ((n+1)B_1 + I)^{-1}$, so $h(B_1) \preceq h(B_2)$. Apply this with $B_1 = X_{c,S}X_{c,S}^\top = \sum_{i\in S} x_{c,i}x_{c,i}^\top \preceq X_{c,T}X_{c,T}^\top = B_2$. For $S = \emptyset$, $M_\emptyset = 0 \preceq M_T$ by (b). □

Corollary 3. *Let $S \subseteq T$, $T \neq \emptyset$, $\lambda_{\min}(G_T) \geq N\varphi_0^2$. Then for every $u \in \mathbb{R}^N$,*

$$\tfrac{n}{n+1}\big[\|(P_T - P_S)u\|^2 - \eta_n\|P_T u\|^2\big] \leq u^\top(M_T - M_S)u \leq \tfrac{n}{n+1}\big[\|(P_T - P_S)u\|^2 + \eta_n\|P_S u\|^2\big].$$

Proof. If $S \neq \emptyset$, Cauchy interlacing gives $\lambda_{\min}(G_S) \geq \lambda_{\min}(G_T) \geq N\varphi_0^2$. Upper bound: $M_T \preceq \frac{n}{n+1}P_T$ and $M_S \succeq (1-\eta_n)\frac{n}{n+1}P_S$ (trivially true for $S = \emptyset$), so $M_T - M_S \preceq \frac{n}{n+1}(P_T - P_S) + \eta_n\frac{n}{n+1}P_S$. Lower bound: $M_T \succeq (1-\eta_n)\frac{n}{n+1}P_T$ and $M_S \preceq \frac{n}{n+1}P_S$. Finally $P_T - P_S$ is the orthogonal projection onto $\operatorname{col}(X_{c,T}) \ominus \operatorname{col}(X_{c,S})$, so $u^\top(P_T - P_S)u = \|(P_T - P_S)u\|^2$. □

Lemma 4 (determinants)**.** $1 \le D_S \le (n+1)^{|S|}$ *for every* S*. Let* $S \subseteq T$ *and* $j = |T| - |S|$*. Then* $D_T/D_S \le (n+1)^j$*, and if* $\lambda_{\min}(G_T) \ge N\varphi_0^2$ *then* $D_T/D_S \ge \big[(n+1)\big(1+\frac{1}{N\varphi_0^2}\big)^{-1}\big]^j$*.*

Proof. With $f(\lambda) = ((n+1)\lambda+1)/(\lambda+1)$, increasing, $f(0)=1$, $f<n+1$, simultaneous diagonalisation gives $D_S = \prod_i f(\lambda_i(G_S))$, whence the first claim. Order eigenvalues increasingly; interlacing gives $\lambda_i(G_T) \le \lambda_i(G_S) \le \lambda_{i+j}(G_T)$, $1 \le i \le |S|$. So $\prod_{i\le|S|} f(\lambda_i(G_T)) \le D_S \le \prod_{i>j} f(\lambda_i(G_T))$. Dividing $D_T = \prod_{i\le|T|} f(\lambda_i(G_T))$ by these: $D_T/D_S \le \prod_{i>|S|} f(\lambda_i(G_T)) \le (n+1)^j$ and $D_T/D_S \ge \prod_{i\le j} f(\lambda_i(G_T)) \ge f(N\varphi_0^2)^j$, with $f(g) \ge (n+1)g/(g+1) = (n+1)(1+1/g)^{-1}$. □

Lemma 5 (prior ratios)**.** *For* $0 \le k \le p-1$*,* $w(k+1)/w(k) = (k+1)/(2p-1-k)$*. Hence:*

(a) $w(k)/w(k') \le (2p)^{k'-k}$ *for* $0 \le k < k' \le p$*;*

(b) for $j \ge 1$ *with* $s_0 + j \le p$*:* $\binom{p-s_0}{j}\frac{w(s_0+j)}{w(s_0)} \le 2\binom{s_0+j}{j}2^{-j} \le 2(j+1)^{s_0}2^{-j}$*.*

Proof. $w(k) = \frac{1}{2}k!(2p-1-k)!/(2p-1)!$ gives the ratio. (a) $w(k)/w(k+1) = (2p-1-k)/(k+1) \le 2p$. (b) Writing both factors as products over $i = 0, \dots, j-1$,

$$\binom{p-s_0}{j}\frac{w(s_0+j)}{w(s_0)} = \prod_{i=0}^{j-1}\frac{p-s_0-i}{i+1}\cdot\frac{s_0+i+1}{2p-s_0-i-1} = \binom{s_0+j}{j}\prod_{i=0}^{j-1}\frac{p-s_0-i}{2p-s_0-i-1}.$$

The denominators are $\ge p$ since $i \le p-s_0-1$. Each factor is $\le \frac{1}{2}$ when $s_0+i \ge 1$ (as $2(p-s_0-i) \le 2p-s_0-i-1$ iff $s_0+i \ge 1$); the only other case is $s_0 = i = 0$, where the factor is $p/(2p-1) \le 1$. So the product is $\le 2\cdot 2^{-j}$. Finally $\binom{s_0+j}{j} = \prod_{i=1}^{s_0}\frac{j+i}{i} \le (j+1)^{s_0}$. □

Lemma 6 (signal left by an underfitting model)**.** *Let* $J = J(S) \ne \emptyset$ *and* $\lambda_{\min}(G_{S'}) \ge N\varphi_0^2$*. Then* $\Delta_S := \|(I-P_S)\mu_c\|^2 \ge N\varphi_0^2\|\theta_J^\star\|^2$*.*

Proof. $\mu_c = X_{c,S_0\cap S}\theta^\star_{S_0\cap S} + X_{c,J}\theta_J^\star$ and $(I-P_S)X_{c,S_0\cap S} = 0$, so $\Delta_S = \min_\gamma \|X_{c,J}\theta_J^\star - X_{c,S}\gamma\|^2$. Since $S' = S \sqcup J$, each vector $X_{c,J}\theta_J^\star - X_{c,S}\gamma$ equals $X_{c,S'}\vartheta$ with $\vartheta = (\theta_J^\star, -\gamma)$, and $\|X_{c,S'}\vartheta\|^2 \ge \lambda_{\min}(G_{S'})\|\vartheta\|^2 \ge N\varphi_0^2\|\theta_J^\star\|^2$. □

Lemma 7 (chi-square facts)**.** *Let* χ_d^2 *be chi-square with* $d \ge 1$ *degrees of freedom.*

(a) For $x > 0$*:* $P(\chi_d^2 \ge d + 2\sqrt{dx} + 2x) \le e^{-x}$ *and* $P(\chi_d^2 \le d - 2\sqrt{dx}) \le e^{-x}$*; moreover* $d + 2\sqrt{dx} + 2x \le 2d + 3x$*. If* $d \le d'$ *then* $P(\chi_d^2 \ge t) \le P(\chi_{d'}^2 \ge t)$ *for all* t*.*

(b) For $\tau > 0$*:* $E\big[e^{\chi_d^2/2}\mathbf{1}\{\chi_d^2 \le \tau\}\big] = \dfrac{(\tau/2)^{d/2}}{\Gamma(d/2+1)} \le \Big(\dfrac{e\tau}{d}\Big)^{d/2}$*.*

Proof. (a) (Standard bounds; proved here in full, so no citation is relied on.) For $0 < \lambda < \frac{1}{2}$, $\log Ee^{\lambda(\chi_d^2-d)} = d(-\lambda - \frac{1}{2}\log(1-2\lambda)) \le d\lambda^2/(1-2\lambda)$, because $-\log(1-u) - u = \sum_{k\ge2}u^k/k \le \frac{u^2}{2}\sum_{k\ge0}u^k = \frac{u^2}{2(1-u)}$ with $u = 2\lambda$. Put $a = \sqrt{d}$, $s = \sqrt{x}$, $t = 2as + 2s^2$ and $\lambda = s/(a+2s) \in (0, \frac{1}{2})$; then $1-2\lambda = a/(a+2s)$, $d\lambda^2/(1-2\lambda) = as^2/(a+2s)$ and $\lambda t = 2s^2(a+s)/(a+2s)$, so Chernoff's bound gives $P(\chi_d^2 - d \ge t) \le \exp\{(as^2 - 2s^2(a+s))/(a+2s)\} = e^{-s^2} = e^{-x}$. For the lower tail, $\log Ee^{-\lambda(\chi_d^2-d)} = d(\lambda - \frac{1}{2}\log(1+2\lambda)) \le d\lambda^2$ for $\lambda > 0$ (as $\log(1+u) \ge u - u^2/2$ for $u \ge 0$), so $P(d - \chi_d^2 \ge t) \le e^{-\lambda t + d\lambda^2} = e^{-t^2/(4d)}$ at $\lambda = t/(2d)$; take $t = 2\sqrt{dx}$. Next $2\sqrt{dx} \le d + x$. Finally $\chi_{d'}^2$ has the law of χ_d^2 plus an independent nonnegative variable.

(b) The density of χ_d^2 is $x^{d/2-1}e^{-x/2}/(2^{d/2}\Gamma(d/2))$, so the truncated moment is

$$\int_0^\tau x^{d/2-1}dx/(2^{d/2}\Gamma(d/2)) = \tau^{d/2}/(2^{d/2}\tfrac{d}{2}\Gamma(\tfrac{d}{2})).$$

And $\Gamma(z+1) \ge \int_z^\infty t^z e^{-t}dt \ge z^z e^{-z}$ for $z > 0$. □

4 The good event

For $J \subseteq [p] \setminus S_0$ with $1 \le |J| \le m_n$ and $s_0 + |J| \le p$ put $\chi^2_J := \sigma_\star^{-2}\|(P_{S_0\cup J} - P_{S_0})e\|^2$. For an underfitting S with $|S| \le m_n$ put $N_S := \|(P_{S'} - P_S)e\|^2$ and $V_S := \|(P_{S'} - P_S)y_c\|^2$. Under SE every $T \subseteq [p]$ with $1 \le |T| \le m_n + s_0$ has $\lambda_{\min}(G_T) \ge N\varphi_0^2 > 0$ (if $m_n + s_0 > p$ this covers every nonempty T), so $X_{c,T}$ has full column rank; in particular $P_{S_0\cup J} - P_{S_0}$ is an orthogonal projection of rank $|J|$ and χ^2_J is exactly $\chi^2_{|J|}$ (Lemma 1(b)), while $\sigma_\star^{-2}N_S$ is $\chi^2_{|J(S)|}$ with $|J(S)| \le s_0$.

Definition 8. $\mathcal{G}_n$ is the intersection of:

(G1) $|\|e\|^2 - N\sigma_\star^2| \le 3\sigma_\star^2\sqrt{N\log n}$;

(G2) $\|P_T e\|^2 \le \sigma_\star^2 C_A(m_n + s_0)\log n$ for every $T \subseteq [p]$ with $1 \le |T| \le m_n + s_0$;

(G3) $\chi^2_J \le \tau_{|J|}$ for every J as above;

(G4) $N_S \le \sigma_\star^2 C_* \log n$ for every underfitting S with $|S| \le m_n$ and $|J'(S)| \le j_*$.

Lemma 9. *Under G, R and SE, $P_0(\mathcal{G}_n^c) \le 5/n$ for $n \ge n_0$.*

Proof. (G1) Lemma 7(a) with $d = N$, $x = \log n$: outside probability $2/n$, $N - 2\sqrt{N\log n} \le \|e\|^2/\sigma_\star^2 \le N + 2\sqrt{N\log n} + 2\log n$, and $2\log n \le \sqrt{N\log n}$ once $N \ge 4\log n$.

(G2) For $|T| = k \ge 1$, $\|P_T e\|^2/\sigma_\star^2$ is χ^2_{rank} with rank $\le k$. There are at most $\binom{p}{k} \le p^k$ such T. With $x_k = k(\log p + 1) + \log n$, Lemma 7(a) and a union bound give exceptional probability $\le p^k e^{-x_k} = e^{-k}/n$; summing over $k \ge 1$ gives $\le 1/n$. On the complement, $\|P_T e\|^2/\sigma_\star^2 \le 2k + 3x_k = 5k + 3k\log p + 3\log n \le (5 + 3A + 3)k\log n = C_A k\log n$, using $\log p \le A\log n$, $\log n \ge 1$, $k \ge 1$; and $k \le m_n + s_0$.

(G3) The same computation for the at most p^j variables χ^2_J with $|J| = j$, each exactly χ^2_j: exceptional probability $\le 1/n$, threshold $C_A j\log n = \tau_j$.

(G4) An underfitting S with $|J'(S)| \le j_*$ is determined by $J'(S)$ (at most $\sum_{j' \le j_*}\binom{p}{j'} \le (j_*+1)p^{j_*}$ choices) and $J(S)$ (a nonempty subset of S_0, fewer than 2^{s_0} choices). Each $N_S/\sigma_\star^2$ is stochastically at most $\chi^2_{s_0}$ (Lemma 7(a), last claim; if $s_0 = 0$ there is no underfitting model). With $x = j_*\log p + s_0\log 2 + \log(j_* + 1) + \log n$, the union bound gives exceptional probability $\le 1/n$, and on the complement $N_S/\sigma_\star^2 \le 2s_0 + 3x \le \big(2s_0 + 3[Aj_* + s_0 + \log(j_* + 1) + 1]\big)\log n = C_*\log n$, using $\log p \le A\log n$ and $1, \log 2 \le \log n$. □

Define the deterministic sequences

$$\delta_n := 3\sqrt{\frac{\log n}{N}} + \frac{C_A(m_n + s_0)\log n}{N}, \quad \delta_n'' := 3\sqrt{\frac{\log n}{N}} + \frac{(12b_0 + 8)(M + \sigma_\star^2)}{N\sigma_\star^2}, \quad \rho_n := \frac{8(M + \sigma_\star^2)}{N\varphi_0^2\sigma_\star^2}.$$

By (2), $\delta_n\log n \to 0$, $\delta_n'' \to 0$, $\rho_n \to 0$.

Lemma 10 (consequences of $\mathcal{G}_n$)**.** *Assume G, R, SE. For $n \ge n_0$, on $\mathcal{G}_n$:*

(a) $\|y_c\|^2 \le 4n(M + \sigma_\star^2)$, and $0 < b \le 6b_0(M + \sigma_\star^2)$.

(b) $\delta_n \le \frac{1}{2}$, and $Q_S^c \ge N\sigma_\star^2(1 - \delta_n)$ for every $S \supseteq S_0$ with $|S| \le m_n + s_0$.

(c) For every underfitting S with $|S| \le m_n$: $2b + Q_S^c \le E_n + V_S$, where $E_n := 2b + \|e\|^2 + \frac{2}{n+1}\|y_c\|^2$ satisfies $N\sigma_\star^2(1 - \delta_n) \le E_n \le N\sigma_\star^2(1 + \delta_n'')$.

(d) If $B \ge N\sigma_\star^2(1 - \delta_n)$, then $\nu\frac{n}{n+1}\frac{\eta_n\|y_c\|^2}{B} \le \rho_n$.

Proof. (a) $\|y_c\|^2 \le 2\|\mu_c\|^2 + 2\|e\|^2 \le 2nM + 2N\sigma_\star^2(1+3\sqrt{\log n/N}) \le 4n(M+\sigma_\star^2)$ once $3\sqrt{\log n/N} \le 1$, by R, Lemma 1(b) and (G1). Then $b = b_0\|y_c\|^2/N \le 4b_0(M+\sigma_\star^2)n/(n-1) \le 6b_0(M+\sigma_\star^2)$; $b > 0$ because $s_y > 0$.

(b) $\delta_n \le \frac{1}{2}$ for $n \ge n_0$ since $\delta_n \to 0$. By Lemma 2(b), $Q_S^c = y_c^\top(I - M_S)y_c \ge y_c^\top(I - P_S)y_c = \|(I-P_S)e\|^2 = \|e\|^2 - \|P_S e\|^2$, using $(I-P_S)\mu_c = 0$. By (G1)–(G2) (for $S = \emptyset$, $\|P_\emptyset e\| = 0$) this is $\ge N\sigma_\star^2 - 3\sigma_\star^2\sqrt{N\log n} - \sigma_\star^2 C_A(m_n + s_0)\log n = N\sigma_\star^2(1-\delta_n)$.

(c) If $S = \emptyset$, $Q_\emptyset^c = \|y_c\|^2 = \|(I-P_\emptyset)y_c\|^2$. If $S \ne \emptyset$, SE gives $\lambda_{\min}(G_S) \ge N\varphi_0^2$, and Lemma 2(c) gives $I - M_S \preceq (I-P_S) + \frac{1+n\eta_n}{n+1}P_S \preceq (I-P_S) + \frac{2}{n+1}P_S$ once $N\varphi_0^2 \ge 1$ (then $n\eta_n \le 1$). In both cases $Q_S^c \le \|(I-P_S)y_c\|^2 + \frac{2}{n+1}\|y_c\|^2$. As $\mathrm{col}(X_{c,S}) \subseteq \mathrm{col}(X_{c,S'})$, $\|(I-P_S)y_c\|^2 = \|(I-P_{S'})y_c\|^2 + V_S$, and $(I-P_{S'})y_c = (I-P_{S'})e$ because $S' \supseteq S_0$; so $\|(I-P_S)y_c\|^2 \le \|e\|^2 + V_S$. This gives the first claim. For the bounds on E_n: $E_n \ge \|e\|^2 \ge N\sigma_\star^2(1-3\sqrt{\log n/N}) \ge N\sigma_\star^2(1-\delta_n)$ by (G1); and by (a) and (G1), $E_n \le 12b_0(M+\sigma_\star^2) + N\sigma_\star^2(1+3\sqrt{\log n/N}) + 8(M+\sigma_\star^2) = N\sigma_\star^2(1+\delta_n'')$.

(d) $\nu = a_0 + n/2 \le n$, $\eta_n\|y_c\|^2 \le 4n(M+\sigma_\star^2)/((n+1)N\varphi_0^2)$ by (a), and $B \ge N\sigma_\star^2/2$. So the left side is at most $8n^3(M+\sigma_\star^2)/((n+1)^2N^2\varphi_0^2\sigma_\star^2) \le \rho_n$, because $n^3 \le (n+1)^2(n-1)$ (equivalent to $n^2 - n - 1 \ge 0$). □

5 Supersets of size at most m_n

For $j \ge 1$ let $\Xi_j := \sum_{J\subseteq[p]\setminus S_0,\,|J|=j} \ell(S_0\cup J)/\ell(S_0)$ ($\Xi_j = 0$ if $s_0 + j > p$). By (1), since every superset $S \in \Omega$ with $|S| \le m_n$ is $S_0 \cup J$ with $1 \le |J| \le m_n$,

$$\Pi(\{S \supsetneq S_0,\ |S| \le m_n\} \mid y) \le \sum_{j=1}^{m_n} \Xi_j. \tag{4}$$

Lemma 11 (one superset)**.** *Assume G, R, SE; $n \ge n_0$. On $\mathcal{G}_n$, for every $J \subseteq [p]\setminus S_0$ with $1 \le |J| = j \le m_n$ and $S = S_0 \cup J$,*

$$\frac{m(S)}{m(S_0)} \le (n+1)^{-j/2}\Big(1 + \frac{1}{N\varphi_0^2}\Big)^{j/2} \exp\Big\{\frac{\upsilon_n}{2}\chi_J^2 + \rho_n\Big\}, \qquad \upsilon_n := \frac{1 + (0.02n+1)/(n^2-1)}{1-\delta_n}.$$

Moreover $1 \le \upsilon_n$, $\upsilon_n - 1 \le 2\big(\delta_n + (0.02n+1)/(n^2-1)\big)$, and $\epsilon_n := (\upsilon_n - 1)C_A\log n/2 \to 0$.

Proof. By (3) (with equal prior weights removed), exactly

$$\frac{m(S)}{m(S_0)} = \Big(\frac{D_{S_0}}{D_S}\Big)^{1/2}\Big(1 + \frac{t_S - t_{S_0}}{2b + Q_S^c}\Big)^{\nu}.$$

Determinant. $|S| = s_0 + j \le m_n + s_0$, so SE and Lemma 4 give $(D_{S_0}/D_S)^{1/2} \le (n+1)^{-j/2}(1 + 1/(N\varphi_0^2))^{j/2}$. *Numerator.* $t_S - t_{S_0} \ge 0$ (Lemma 2(d)); Corollary 3 with $(S,T) = (S_0, S)$ and $u = y_c$, together with $(P_S - P_{S_0})\mu_c = 0$ (both projections fix μ_c), gives $t_S - t_{S_0} \le \frac{n}{n+1}[\sigma_\star^2\chi_J^2 + \eta_n\|y_c\|^2]$. *Denominator.* $2b + Q_S^c \ge Q_S^c \ge N\sigma_\star^2(1-\delta_n)$ by Lemma 10(b). With $\log(1+x) \le x$ and Lemma 10(d) (with $B = 2b + Q_S^c$),

$$\nu\log\Big(1 + \frac{t_S - t_{S_0}}{2b + Q_S^c}\Big) \le \frac{\nu n}{(n+1)N(1-\delta_n)}\chi_J^2 + \rho_n,$$

and $\frac{\nu n}{(n+1)N} = \frac{(0.01+n/2)n}{n^2-1} = \frac{1}{2}(1 + \frac{0.02n+1}{n^2-1})$, so the first term is $\frac{\upsilon_n}{2}\chi_J^2$. Finally $\upsilon_n \ge 1$ is clear, $\upsilon_n - 1 = \frac{(0.02n+1)/(n^2-1) + \delta_n}{1-\delta_n}$ with $1 - \delta_n \ge \frac{1}{2}$, and $\epsilon_n \to 0$ because $\delta_n \log n \to 0$ and $\log n/n \to 0$. □

Proposition 12 (superset sums)**.** *Assume G, R, SE. Define the nonnegative random variable*

$$\Upsilon_n := \sum_{j=1}^{m_n} \sum_{\substack{J\subseteq[p]\setminus S_0\\ |J|=j}} \frac{w(s_0+j)}{w(s_0)}(n+1)^{-j/2}\Big(1+\frac{1}{N\varphi_0^2}\Big)^{j/2} e^{j\epsilon_n}\, e^{\chi_J^2/2}\,\mathbf{1}\{\chi_J^2\le\tau_j\},$$

and let $\Upsilon_n^{>j_1}$ be the part with $j>j_1$. Put

$$q_n := \Big[\frac{eC_A\log n}{4(n+1)}\Big(1+\frac{1}{N\varphi_0^2}\Big)e^{2\epsilon_n}\Big]^{1/2}, \qquad K_{s_0,j_1} := \sum_{i\ge 0}(j_1+2+i)^{s_0}2^{-i}.$$

Then for $n\ge n_0$: (a) on $\mathcal{G}_n$, $\sum_{j=j_1+1}^{m_n}\Xi_j \le e^{\rho_n}\Upsilon_n^{>j_1}$ for every integer $j_1\ge 0$; (b) $q_n\le\min\{\frac12,\sqrt{eC_A}\,r_n\}$ and $E_0\Upsilon_n^{>j_1}\le 2K_{s_0,j_1}q_n^{j_1+1}$; (c) $\Upsilon_n = O_{P_0}(r_n)$ and $\sum_{j\le m_n}\Xi_j\mathbf{1}_{\mathcal{G}_n} = O_{P_0}(r_n)$.

Proof. (a) On $\mathcal{G}_n$, (G3) makes every indicator in Υ_n equal to one, and for $\chi_J^2\le\tau_j$, $e^{\upsilon_n\chi_J^2/2}\le e^{\chi_J^2/2}e^{(\upsilon_n-1)\tau_j/2} = e^{\chi_J^2/2}e^{j\epsilon_n}$ (as $\upsilon_n\ge 1$). Insert into Lemma 11, multiply by $w(s_0+j)/w(s_0)$ and sum over J and $j>j_1$.

(b) $q_n^2 = \frac{eC_A\log n}{n}\cdot\frac{n}{4(n+1)}(1+\frac{1}{N\varphi_0^2})e^{2\epsilon_n}$ and the second factor tends to $\frac14$, so $q_n\le\sqrt{eC_A}r_n$ and $q_n\le\frac12$ for $n\ge n_0$. All weights in Υ_n are deterministic and each χ_J^2 is exactly χ_j^2, so by Lemma 7(b) each term has expectation at most the weight times $(e\tau_j/j)^{j/2} = (eC_A\log n)^{j/2}$. By Lemma 5(b),

$$E_0\Upsilon_n^{>j_1} \le \sum_{j>j_1} 2(j+1)^{s_0}2^{-j}(n+1)^{-j/2}\Big(1+\frac{1}{N\varphi_0^2}\Big)^{j/2}e^{j\epsilon_n}(eC_A\log n)^{j/2} = \sum_{j>j_1}2(j+1)^{s_0}q_n^j,$$

and with $q_n\le\frac12$, $\sum_{j>j_1}(j+1)^{s_0}q_n^j = q_n^{j_1+1}\sum_{i\ge0}(j_1+2+i)^{s_0}q_n^i\le K_{s_0,j_1}q_n^{j_1+1}$.

(c) By (b) with $j_1=0$ and Markov's inequality, $P_0(\Upsilon_n>2LK_{s_0,0}q_n)\le 1/L$ for all $L>0$, and $q_n\le\sqrt{eC_A}r_n$; so $\Upsilon_n = O_{P_0}(r_n)$. By (a), $\sum_{j\le m_n}\Xi_j\mathbf{1}_{\mathcal{G}_n}\le e^{\rho_n}\Upsilon_n$ with $\rho_n\to 0$. □

6 Underfitting models of size at most m_n

Throughout, S is underfitting with $|S|\le m_n$, $J=J(S)$, $J'=J'(S)$, $|S'|=|S|+|J|\le m_n+s_0$; so SE applies to S (if nonempty) and to S'. The model S need not lie in Ω for the bounds of this section.

Lemma 13 (S against S')**.** *Assume G, R, SE; $n\ge n_0$. On $\mathcal{G}_n$, for every such S,*

$$\begin{aligned}\frac{\ell(S)}{\ell(S')} &\le (2p)^{|J|}(n+1)^{|J|/2}\exp\Big\{-\nu\frac{n}{n+1}\frac{V_S}{E_n+V_S}+\rho_n\Big\}\\ &\le (2p)^{|J|}(n+1)^{|J|/2}\exp\Big\{-\min\Big(\frac{V_S}{8\sigma_\star^2},\frac{n}{5}\Big)+\rho_n\Big\}.\end{aligned}$$

In particular $\ell(S)/\ell(S')\le(2p)^{s_0}(n+1)^{s_0/2}e^{\rho_n}$.

Proof. Prior. $|S'|=|S|+|J|\le p$, so $w(|S|)/w(|S'|)\le(2p)^{|J|}$ by Lemma 5(a). *Determinant.* $(D_{S'}/D_S)^{1/2}\le(n+1)^{|J|/2}$ by Lemma 4. *Variance factor.* By (3) the remaining factor is $(\frac{2b+Q_{S'}^c}{2b+Q_S^c})^\nu = (1-x)^\nu$ with $x:=\frac{t_{S'}-t_S}{2b+Q_S^c}$. Here $x\ge0$ by Lemma 2(d), and $x<1$ since $2b+Q_{S'}^c>0$. So $(1-x)^\nu\le e^{-\nu x}$. By Corollary 3 (pair $S\subseteq S'$, $u=y_c$) and $\|P_{S'}y_c\|\le\|y_c\|$, $t_{S'}-t_S\ge\frac{n}{n+1}[V_S-\eta_n\|y_c\|^2]$; by Lemma 10(c), $0<2b+Q_S^c\le E_n+V_S$. Hence

$$\nu x\ge\nu\frac{n}{n+1}\frac{V_S-\eta_n\|y_c\|^2}{E_n+V_S}\ge\nu\frac{n}{n+1}\frac{V_S}{E_n+V_S}-\nu\frac{n}{n+1}\frac{\eta_n\|y_c\|^2}{E_n}\ge\nu\frac{n}{n+1}\frac{V_S}{E_n+V_S}-\rho_n.$$

(First step: if $V_S - \eta_n\|y_c\|^2 \ge 0$, enlarging the denominator decreases the quotient; otherwise the right side is negative while $\nu x \ge 0$. Second step: $E_n + V_S \ge E_n > 0$. Third: Lemma 10(c),(d) with $B = E_n$.) This is the first inequality. For the second, $\frac{V}{E_n+V} \ge \min(\frac{V}{2E_n}, \frac{1}{2})$. With $\nu \ge n/2$, $E_n \le N\sigma_\star^2(1+\delta_n'')$ and $\delta_n'' \le 1$ ($n \ge n_0$): $\nu\frac{n}{n+1}\frac{V_S}{2E_n} \ge \frac{n^2}{n^2-1}\cdot\frac{V_S}{4\sigma_\star^2(1+\delta_n'')} \ge \frac{V_S}{8\sigma_\star^2}$, and $\nu\frac{n}{n+1}\cdot\frac{1}{2} \ge \frac{n^2}{4(n+1)} \ge \frac{n}{5}$ for $n \ge 4$. The last claim uses $V_S \ge 0$, $|J| \le s_0$ and $2p, n+1 \ge 1$. □

Lemma 14 (adding the signal)**.** *Assume G, R, SE; $n \ge n_0$. On $\mathcal{G}_n$, if in addition $|J'(S)| \le j_*$ and $\Delta_S \ge L\sigma_\star^2 \log n$ for some $L \ge 4C_*$, then*

$$\frac{\ell(S)}{\ell(S')} \le \exp\Big\{s_0(A+1)\log n + \rho_n - \min\Big(\frac{L\log n}{32}, \frac{n}{5}\Big)\Big\}.$$

Proof. $P_{S'}\mu_c = \mu_c$, so $(P_{S'} - P_S)y_c = (I - P_S)\mu_c + (P_{S'} - P_S)e$ and $\sqrt{V_S} \ge \sqrt{\Delta_S} - \sqrt{N_S}$. By (G4), $N_S \le \sigma_\star^2 C_* \log n \le \frac{C_*}{L}\Delta_S \le \frac{1}{4}\Delta_S$, so $V_S \ge \frac{1}{4}\Delta_S \ge \frac{L}{4}\sigma_\star^2 \log n$ and $V_S/(8\sigma_\star^2) \ge L\log n/32$. For the prefactor: $|J| \le s_0$, $p \le n^A$, and $\log 2 + \frac{1}{2}\log(n+1) \le \log n$ for $n \ge 5$, so $(2p)^{|J|}(n+1)^{|J|/2} \le e^{s_0(A+1)\log n}$. Apply Lemma 13. □

Lemma 15 (assembling the underfitting mass)**.** *Assume G, R, SE; $n \ge n_0$. Let $\mathcal{U}$ be a collection of underfitting models with $|S| \le m_n$ and*

$$\kappa_n(\mathcal{U}) := \max\Big\{\frac{\ell(S)}{\ell(S')} : S \in \mathcal{U},\ |J'(S)| \le j_*\Big\} \quad (\max\emptyset := 0).$$

On $\mathcal{G}_n$,

$$\sum_{S\in\mathcal{U}} \frac{\ell(S)}{\ell(S_0)} \le 2^{s_0}\kappa_n(\mathcal{U})(1 + e^{\rho_n}\Upsilon_n) + 2^{s_0}(2p)^{s_0}(n+1)^{s_0/2}e^{2\rho_n}\Upsilon_n^{>j_*}, \tag{5}$$

and $(2p)^{s_0}(n+1)^{s_0/2}\Upsilon_n^{>j_*} = O_{P_0}\big((\log n)^{(j_*+1)/2}/n\big)$.

Proof. Write $\frac{\ell(S)}{\ell(S_0)} = \frac{\ell(S)}{\ell(S')}\cdot\frac{\ell(S')}{\ell(S_0)}$. A given $S' = S_0 \cup J'$ arises from at most $2^{s_0} - 1$ underfitting S, namely $S = J' \cup (S_0 \setminus I)$ with $\emptyset \ne I \subseteq S_0$. Note $|J'(S)| \le |S| \le m_n$. For $|J'| \le j_*$ bound the first factor by $\kappa_n(\mathcal{U})$; the second factor, summed over the distinct S', is 1 for $S' = S_0$ plus terms of $\sum_{j=1}^{m_n}\Xi_j \le e^{\rho_n}\Upsilon_n$ (Proposition 12(a), $j_1 = 0$). For $|J'| > j_*$ bound the first factor by $(2p)^{s_0}(n+1)^{s_0/2}e^{\rho_n}$ (Lemma 13) and the sum of the second over distinct S' by $\sum_{j=j_*+1}^{m_n}\Xi_j \le e^{\rho_n}\Upsilon_n^{>j_*}$. This is (5).

By Proposition 12(b) and $q_n^2 \le eC_A\log n/n$, $E_0[(2p)^{s_0}(n+1)^{s_0/2}\Upsilon_n^{>j_*}] \le 2K_{s_0,j_*}2^{s_0}n^{As_0}(n+1)^{s_0/2}(eC_A\log n)^{(j_*+1)/2}n^{-(j_*+1)/2}$. Since $j_* \ge (2A+1)s_0 + 1$, $\frac{j_*+1}{2} \ge s_0(A+\frac{1}{2}) + 1$, so $n^{As_0}(n+1)^{s_0/2}n^{-(j_*+1)/2} \le (1+\frac{1}{n})^{s_0/2}n^{-1} \le 2^{s_0/2}n^{-1}$. Markov's inequality gives the O_{P_0} claim. □

7 Proof of the theorem

In (i) and (iv), and in Steps 0–1, $n \ge n_0$; the deterministic parts of (ii) and (iii) hold for every n (see there). Three elementary facts are used repeatedly. First, a finite sum of $O_{P_0}(a_n)$ terms is $O_{P_0}(a_n)$. Second, a bound valid for $n \ge n_0$, or of the form $\lim_{L\to\infty}\limsup_n P_0(X_n > La_n) = 0$, gives tightness of X_n/a_n over all n, because for each fixed n the single finite random variable X_n/a_n is tight ($a_n > 0$), and a finite family of tight variables is tight; for the posterior masses bounded below this is explicit, since they lie in $[0,1]$ and so $X_n/a_n \le 1/a_n$. Third, if $Y_n \ge 0$, $Y_n\mathbf{1}_{\mathcal{G}_n} = O_{P_0}(a_n)$ and $P_0(\mathcal{G}_n^c) \to 0$, then $Y_n = O_{P_0}(a_n)$, because $P_0(Y_n > La_n) \le P_0(Y_n\mathbf{1}_{\mathcal{G}_n} > La_n) + P_0(\mathcal{G}_n^c)$. Also $n^{-2} \le n^{-1/2} \le r_n$ and $(\log n)^{(j_*+1)/2}/n = o(r_n)$.

Step 0: the tail above m_n (H-T02). By H-T02 with $m = m_n$, and $2^{-m_n} \le (2p)^{-s_0} n^{-(s_0+1)/2} n^{-1/2}$ (definition of m_n),

$$\Pi(|S| > m_n \mid y) \le (2p)^{-s_0} n^{-(s_0+2)/2} (2p)^{s_0} (n+1)^{s_0/2} \sqrt{n}\, \Theta_{n,m_n} = (1+\tfrac{1}{n})^{s_0/2} n^{-1/2} \Theta_{n,m_n}, \quad (6)$$

and $P_0(\Theta_{n,m_n} > L) \le \sup_m P_0(\Theta_{n,m} > L)$, so $\Pi(|S| > m_n \mid y) = O_{P_0}(n^{-1/2})$.

Step 1: the signal under BM-C. Let $\mathcal{U}$ be the set of underfitting $S \in \Omega$ with $|S| \le m_n$. For $S \in \mathcal{U}$, Lemma 6 (SE on S') and BM-C give

$$\Delta_S \ge N\varphi_0^2 \|\theta^\star_{J(S)}\|^2 \ge N\varphi_0^2 \min_{j \in S_0} \theta_j^{\star 2} \ge \Big(1 - \frac{1}{n}\Big) M_* \sigma_\star^2 \log n \ge L_* \sigma_\star^2 \log n,$$

since $J(S) \ne \emptyset$ and $(1 - \frac{1}{n})(2L_* + 1) \ge L_*$ for $n \ge 2$. As $L_* \ge 4C_*$, $L_*/32 \ge s_0(A+1) + 2$, and $n/5 \ge (s_0(A+1)+2)\log n$ for $n \ge n_0$, Lemma 14 with $L = L_*$ gives on $\mathcal{G}_n$

$$\kappa_n(\mathcal{U}) \le \exp\{s_0(A+1)\log n + \rho_n - (s_0(A+1)+2)\log n\} = e^{\rho_n} n^{-2}. \quad (7)$$

Under BM-∞, $M_n \ge M_*$ for all large n, so BM-C holds for all large n; since the conclusions are asymptotic, BM-∞ is covered by the BM-C case. (If $s_0 = 0$, $\mathcal{U} = \emptyset$ and Step 1 is void.)

(i). Every $S \in \Omega \setminus \{S_0\}$ either has $|S| > m_n$, or is a superset with $|S| \le m_n$, or lies in $\mathcal{U}$ (if $J(S) = \emptyset$ and $S \ne S_0$ then $J'(S) \ne \emptyset$). Hence

$$1 - \pi(S_0) \le \Pi(|S| > m_n \mid y) + \Pi(\{S \supsetneq S_0, |S| \le m_n\} \mid y) + \Pi(\mathcal{U} \mid y).$$

The first term is $O_{P_0}(n^{-1/2})$ by (6). The second is at most $\sum_{j \le m_n} \Xi_j$ by (4), which is $O_{P_0}(r_n)$ on $\mathcal{G}_n$ by Proposition 12(c). By (1), (5) and (7), on $\mathcal{G}_n$,

$$\begin{aligned}\Pi(\mathcal{U} \mid y) \le \sum_{S \in \mathcal{U}} \frac{\ell(S)}{\ell(S_0)} &\le 2^{s_0} e^{\rho_n} n^{-2} (1 + e^{\rho_n} \Upsilon_n) + 2^{s_0} e^{2\rho_n} (2p)^{s_0} (n+1)^{s_0/2} \Upsilon_n^{> j_*} \\ &= O_{P_0}\Big(n^{-2} + \frac{(\log n)^{(j_*+1)/2}}{n}\Big),\end{aligned}$$

using $\Upsilon_n = O_{P_0}(r_n)$ and Lemma 15. Since $P_0(\mathcal{G}_n^c) \le 5/n$ (Lemma 9), all three terms are $O_{P_0}(r_n)$ without restriction to $\mathcal{G}_n$, and $1 - \pi(S_0) = O_{P_0}(r_n)$.

(ii). For $j \in S_0$, $1 - \mathrm{PIP}_j = \pi(\{S : j \notin S\}) \le \pi(\Omega \setminus \{S_0\}) = 1 - \pi(S_0)$. For $j \notin S_0$, $\mathrm{PIP}_j = \pi(\{S : j \in S\}) \le 1 - \pi(S_0)$. This uses only $S_0 \in \Omega$, so it holds for every data set and every n with $s_0 \le K$; when $s_0 > K$, $\pi(S_0) := 0$ (reading 2 of `STATEMENT.md`) and the bound $|\mathrm{PIP}_j - \mathbf{1}\{j \in S_0\}| \le 1$ is trivial. On the event $\{1 - \pi(S_0) < \frac{1}{2}\}$ it follows that $\mathrm{PIP}_j > \frac{1}{2}$ exactly for $j \in S_0$, i.e. $\mathrm{MPM} = S_0$. By (i), $1 - \pi(S_0) \to 0$ in P_0-probability, so $P_0(\mathrm{MPM} = S_0) \ge P_0(1 - \pi(S_0) < \frac{1}{2}) \to 1$.

(iii). On $\{\pi(S_0) > \frac{1}{2}\}$, every $S \in \Omega \setminus \{S_0\}$ has $\pi(S) \le 1 - \pi(S_0) < \frac{1}{2} < \pi(S_0)$, i.e. $F(S) < F(S_0)$: S_0 is the strict global maximiser of F on Ω, so $S^\dagger = S_0$ and the tie rule is not invoked. Hence $\{S^\dagger \ne S_0\} \subseteq \{1 - \pi(S_0) \ge \frac{1}{2}\}$ and by Markov's inequality $P_0(S^\dagger \ne S_0) \le 2E_0[1 - \pi(S_0)]$. This argument holds for every n with $s_0 \le K$. (For the finitely many n with $s_0 > K$, $S_0 \notin \Omega$, $\pi(S_0) := 0$ and the bound reads $P_0(S^\dagger \ne S_0) \le 2$.) Finally, since $0 \le 1 - \pi(S_0) \le 1$, for every $\epsilon \in (0,1)$, $E_0[1 - \pi(S_0)] \le \epsilon + P_0(1 - \pi(S_0) > \epsilon)$, and the last probability tends to 0 by (i); so $\limsup_n E_0[1 - \pi(S_0)] \le \epsilon$ for every ϵ, i.e. $E_0[1 - \pi(S_0)] \to 0$.

(iv). Here BM-C and BM-∞ are not assumed. If $J_n = \emptyset$, then $\Pi(\{S \not\supseteq J_n\} \mid y) = 0$ and there is nothing to prove. Otherwise, for $j \in J_n$ let $\mathcal{U}_j$ be the set of $S \in \Omega$ with $|S| \le m_n$ and $j \notin S$. Each $S \in \mathcal{U}_j$ is underfitting ($j \in J(S)$), and by Lemma 6 and the definition of J_n,

$$\Delta_S \ge N\varphi_0^2 \|\theta^\star_{J(S)}\|^2 \ge N\varphi_0^2 \theta_j^{\star 2} \ge \Big(1 - \frac{1}{n}\Big) M_* \sigma_\star^2 \log n \ge L_* \sigma_\star^2 \log n.$$

No condition on the other coordinates of $\theta^\star$ is used: they are all fitted by $S' = S \cup S_0$. Exactly as in Step 1, $\kappa_n(\mathcal{U}_j) \le e^{\rho_n} n^{-2}$ on $\mathcal{G}_n$, and (5) gives, on $\mathcal{G}_n$, a bound on $\Pi(\mathcal{U}_j \mid y)$ that does not depend on j:

$$\Pi(\mathcal{U}_j \mid y) \le 2^{s_0} e^{\rho_n} n^{-2} (1 + e^{\rho_n} \Upsilon_n) + 2^{s_0} e^{2\rho_n} (2p)^{s_0} (n+1)^{s_0/2} \Upsilon_n^{> j_*}.$$

Since $\{S \in \Omega : S \not\supseteq J_n\} \subseteq \{|S| > m_n\} \cup \bigcup_{j \in J_n} \mathcal{U}_j$ and $|J_n| \le s_0$,

$$\begin{aligned} \Pi(\{S \not\supseteq J_n\} \mid y) &\le \Pi(|S| > m_n \mid y) + s_0 \max_{j \in J_n} \Pi(\mathcal{U}_j \mid y) \\ &= O_{P_0}(n^{-1/2}) + O_{P_0}\Big(n^{-2} + \frac{(\log n)^{(j_*+1)/2}}{n}\Big) = O_{P_0}(r_n), \end{aligned}$$

by (6), Lemma 15 and Lemma 9. For $j \in J_n$, $1 - \mathrm{PIP}_j = \Pi(\{S : j \notin S\} \mid y) \le \Pi(\{S \not\supseteq J_n\} \mid y)$, so $1 - \min_{j \in J_n} \mathrm{PIP}_j = O_{P_0}(r_n)$ and $\min_{j \in J_n} \mathrm{PIP}_j \to 1$ in P_0-probability.

(v). If $s_0 = 0$, the minimum in BM-C and BM-∞ is over the empty set, hence $+\infty$, and both conditions hold. For $c > 0$, the BM-C inequality for $(\theta^\star/c, \sigma_\star^2/c^2)$ reads $\min_j \theta_j^{\star 2}/c^2 \ge M_*(\sigma_\star^2/c^2) \log n/(n\varphi_0^2)$, which is the original inequality divided by c^2; φ_0 depends only on the standardised design and M_*, n do not involve the scale. The same holds for BM-∞ and for each defining inequality of J_n. This applies in particular to $c = s_y$. □

8 Remarks

Remark 16 (on the constant M_*)**.** BM-C enters only through Step 1 and (iv), via $(1 - \frac{1}{n})M_* \ge L_*$. The proof therefore gives (i)–(iv) with M_* replaced by any constant M with $M > L_*$ (then $(1 - \frac{1}{n})M \ge L_*$ for large n). This is a statement about this proof, not about sharpness; no lower bound on the needed signal is proved here.

Remark 17 (where the hypotheses enter)**.** G gives the exact Gaussian law of e (Lemma 1). R gives $p \le n^A$ (all union bounds and j_*), $\|\mu\|^2 \le nM$ (Lemma 10) and $s_0 \le K$ (so $S_0 \in \Omega$). SE at sizes up to $m_n + s_0$ gives full rank and the eigenvalue bounds for S, S' and $S_0 \cup J$. H-T02 is used only in Step 0. The inverse-gamma rate $b = b_0 s_y^2$ is random but common to all models; it enters only through $2b \ge 0$ and the bound $b \le 6b_0(M + \sigma_\star^2)$ on $\mathcal{G}_n$, so no change-of-rate argument is needed for the comparisons made here. The comparison of an underfitting S with $S' = S \cup S_0$ (whose residual is pure noise) is what makes the beta-min scale $\sigma_\star^2$ rather than $\sigma_\star^2 + M$.

Remark 18 (the cap)**.** All comparisons are written with $\ell(S) = w(|S|)\, m(S)$, which is defined for every $S \subseteq [p]$; models S' or $S_0 \cup J$ outside Ω enter only as upper-bound devices, through nonnegative sums. The cap is used only to place S_0 in Ω.